

Property Tables of Superheated Steam

T °C	v m ³ /kg	u kJ/kg	h kJ/kg	s kJ/kg·K
$P = 1.00 \text{ MPa (179.88°C)}$				
179.88	0.19437	2582.8	2777.1	6.5850
200	0.20602	2622.3	2828.3	6.6956
250	0.23275	2710.4	2943.1	6.9265
300	0.25799	2793.7	3051.6	7.1246
350	0.28250	2875.7	3158.2	7.3029
400	0.30661	2957.9	3264.5	7.4670
500	0.35411	3125.0	3479.1	7.7642
600	0.40111	3297.5	3698.6	8.0311
700	0.44783	3476.3	3924.1	8.2755
800	0.49438	3661.7	4156.1	8.5024
900	0.54083	3853.9	4394.8	8.7150
1000	0.58721	4052.7	4640.0	8.9155
1100	0.63354	4257.9	4891.4	9.1057
1200	0.67983	4469.0	5148.9	9.2866
1300	0.72610	4685.8	5411.9	9.4593

Find least squares regression of enthalpy (h) as functions of temperature (T) using polynomials of order 3 and 4.

$$T_{sat} = 179.88^\circ\text{C}, h_{sat} = 2777.1 \text{ kJ/kg}$$

$$x = \frac{T - T_{sat}}{1000}, y = \frac{h - h_{sat}}{1000}$$

i	$x_i = T_i - T_{sat}$	$y_i = h_i - h_{sat}$
0	0	0
1	0.02012	0.0512
2	0.07012	0.1660
$\vdots$	$\vdots$	$\vdots$

$$y(x) = \Phi_3(x, \mathbf{a}) = a_1x + a_2x^2 + a_3x^3$$

$$y(x) = \Phi_4(x, \mathbf{a}) = a_1x + a_2x^2 + a_3x^3 + a_4x^4$$